

TELECOM AGENT PORTFOLIO

RAG Customer Care Assistant for Mobile Service Operations

LangChain RAG · Chroma Multi-Source Retrieval · Groq Qwen3-32B · Streamlit

3

Knowledge Stores

9

Context Docs / Query

1

Unified RAG Chain

2

User Channels

Portfolio Project | Telecom Support Automation | 2026

Build a More Reliable Telecom Support Experience

Challenge

Support teams handle repetitive billing, roaming, SIM and network tickets with inconsistent resolution quality.

Approach

A retrieval-augmented assistant grounds every answer in curated FAQ, resolved tickets, and telecom guide content.

Architecture

Merged top-k retrieval from 3 Chroma collections feeds a deterministic prompt to Qwen3-32B on Groq.

Outcome

Faster first response, stronger answer consistency, and transparent source-backed troubleshooting steps.

Telecom Care Requires Context, Not Generic Chat

Current Pain Points

Fragmented knowledge across FAQ pages, ticket history, and long PDF guides

High repeated volume for common incidents (slow data, call drops, roaming charges)

Agent responses can drift without evidence-backed retrieval

Escalations increase when first-line answers lack actionable steps

Design Principles

Grounded responses

Answer strictly from retrieved telecom context

Multi-source evidence

Blend FAQ, ticket resolution patterns, and guide chunks

Deterministic behavior

Temperature 0 and explicit system policy reduce variance

Operational usability

Serve both web support and terminal support workflows

SOLUTION

Unified Telecom RAG Assistant Across Channels

Question Intake

Customer asks in Streamlit chat or CLI shell.

Merged Retrieval

Top-3 from each collection returns 9 grounded context documents.

Prompted Reasoning

Structured system prompt aligns behavior to telecom support scope.

Actionable Answer

Response explains likely cause and step-by-step next actions.

Fallback Path

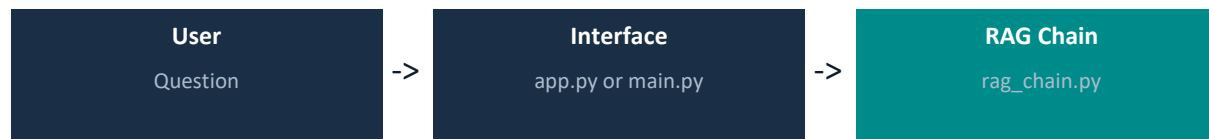
If context is weak, assistant advises 611 or MyTelecom app.

Extensible Stack

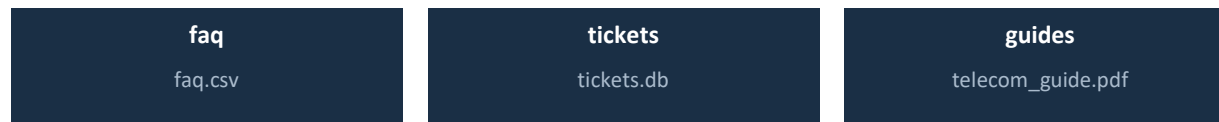
Ingestion scripts allow refreshing knowledge when policies change.

Evidence Flow from User Query to Reliable Answer

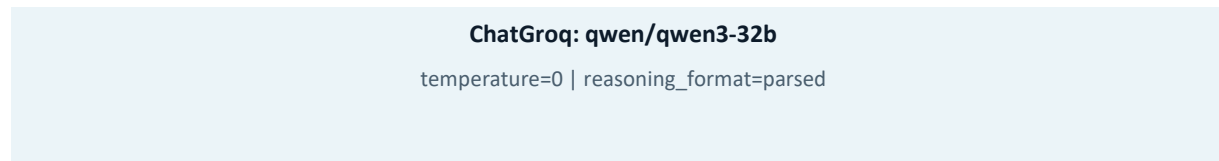
Request Pipeline



Retriever Layer



Model Layer



Why This Works

Three complementary sources reduce single-source blind spots

Embedding model is lightweight and local: all-MiniLM-L6-v2

Merged retrieval preserves domain breadth for each query

Strict prompting keeps responses professional and bounded

Three Data Pipelines Power the Assistant

Collection	Source	Ingestion	Granularity	Role in Answer
faq	data/faq.csv	ingest_faq.py	1 row = 1 doc	Policy and how-to guidance
tickets	data/tickets.db	ingest_tickets.py	1 ticket = 1 doc	Real incident-resolution patterns
guides	data/telecom_guide.pdf	ingest_pdf.py	600/100 chunks	Detailed technical reference

Retriever policy: top-3 from each collection -> 9 context documents per query, then formatted with source metadata before LLM invocation.

Consistent Answer Engine Across Web and CLI

Streamlit Journey

User selects sample question or types custom issue

`build_chain()` is cached once per session

`chain.stream(question)` returns incremental answer tokens

Conversation state persists in `st.session_state`

CLI Journey

Operator starts `main.py` interactive shell

User enters issue text; quit/exit supported

Same `chain.stream(question)` prints answer progressively

Uniform logic preserves parity with web responses

Quality and Operational Guardrails

Grounding Constraint

System prompt forces answers to rely only on retrieved context and avoid unsupported claims.

Deterministic Inference

temperature=0, bounded retry policy, and stable prompt structure improve repeatability.

Safe Fallback

If evidence is insufficient, assistant explicitly recommends 611 or MyTelecom app escalation.

Refreshable Knowledge

Independent ingest scripts allow controlled updates as FAQ policy or guide content changes.

From Portfolio Prototype to Production-Ready Telecom Assistant

Phase 1	Phase 2	Phase 3	Phase 4
Knowledge hardening Add structured citation output and confidence scoring	Channel expansion Integrate WhatsApp/web widget and CRM context handoff	Ops analytics Track intent mix, retrieval hit-rate, and escalation reasons	Automation Add workflow tools for SIM reset and roaming pack checks

Expected Business Impact

- Lower average handle time for repetitive telecom support intents
- Higher first-contact resolution through source-grounded instructions
- Reduced response variance across channels and support agents
- Clear path to monitored, auditable AI-assisted operations

Telecom RAG Agent

A practical blueprint for reliable AI customer support

REPOSITORY

rag-telecom-chatbot (Streamlit + LangChain + Chroma + Groq)

Core files: app.py | rag_chain.py | retriever.py | ingest_*.py

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Collections

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Docs/Query

openai/gpt-oss-120b

LLM

Python 3.11+

Runtime

Built for telecom operations: faster support, safer responses, clearer escalation.